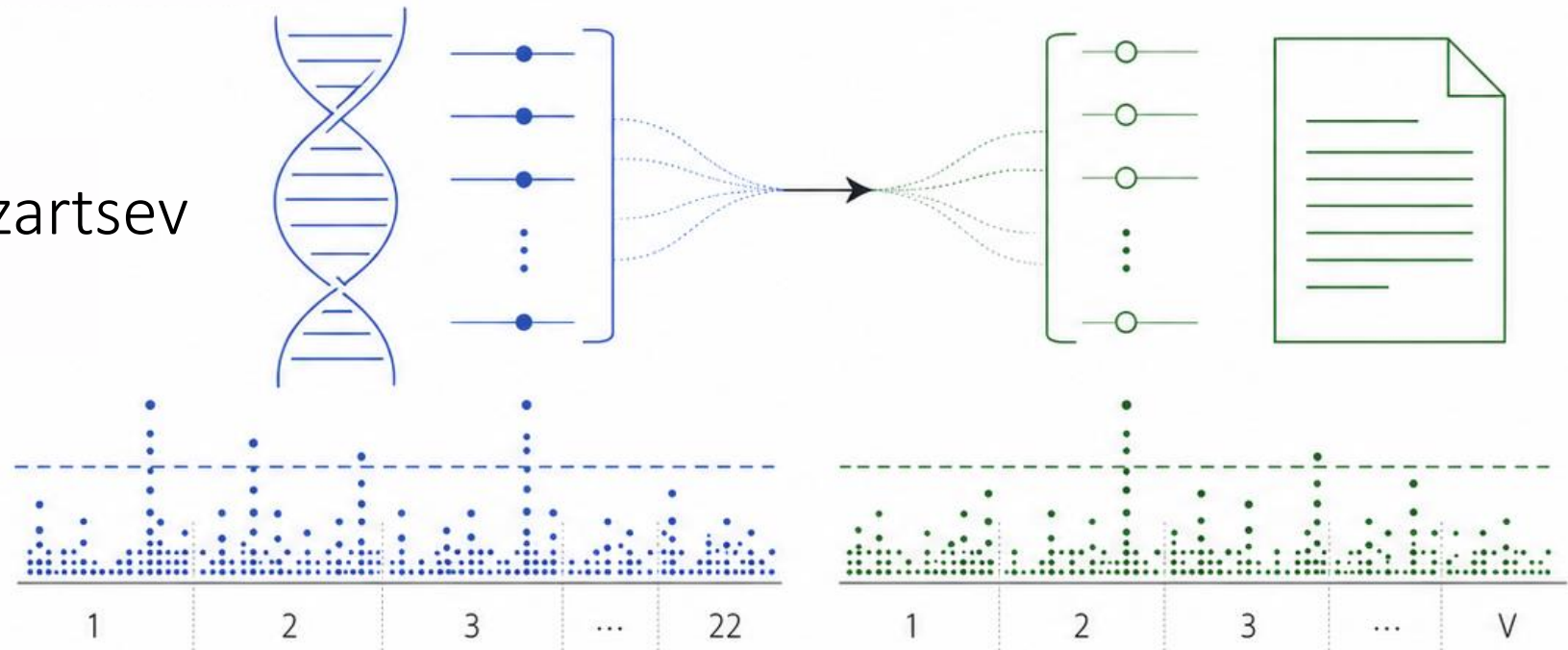


From Genes to Tokens:

A GWAS-Inspired Approach for Interpretable Stylometric Analysis



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what is stylometry?

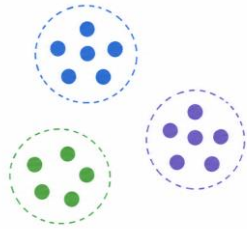
Stylometry

the computational analysis of linguistic style
based on measurable features of texts

Basic assumption

authors have stable linguistic habits
that appear unconsciously in texts

stylometry tasks



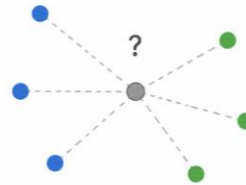
clusterization

data

n texts

question

how many distinctive groups (hopefully author groups) in the dataset?



classification

data

$n+1$ texts by k authors

question

who is the author of $n+1$ th text if the authors of n texts are known?



interpretation

data

n texts by k authors

question

which features are the distinctive markers of known groups?

roadmap: from attribution to interpretable stylometry

1960s: statistical authorship attribution

Function words and probabilistic models show that authorship can be inferred from stable linguistic habits.

2002: Burrows's Delta

Texts are compared through standardized (z-score) frequencies of the most frequent words. Delta is transparent in principle, but its main output is still an aggregate distance.

2007–2010s: specific vocabulary, characteristic features

Stylometry moves toward interpretation: researchers identify words that are overused, underused, or characteristic of particular authors and groups.

2016–2017: explaining Delta

Delta measures are decomposed into feature selection, scaling, and distance functions. The field becomes more methodologically transparent.

2020s: neural authorship attribution

DL and ML increase accuracy, but explanations often rely on model-dependent feature attribution methods

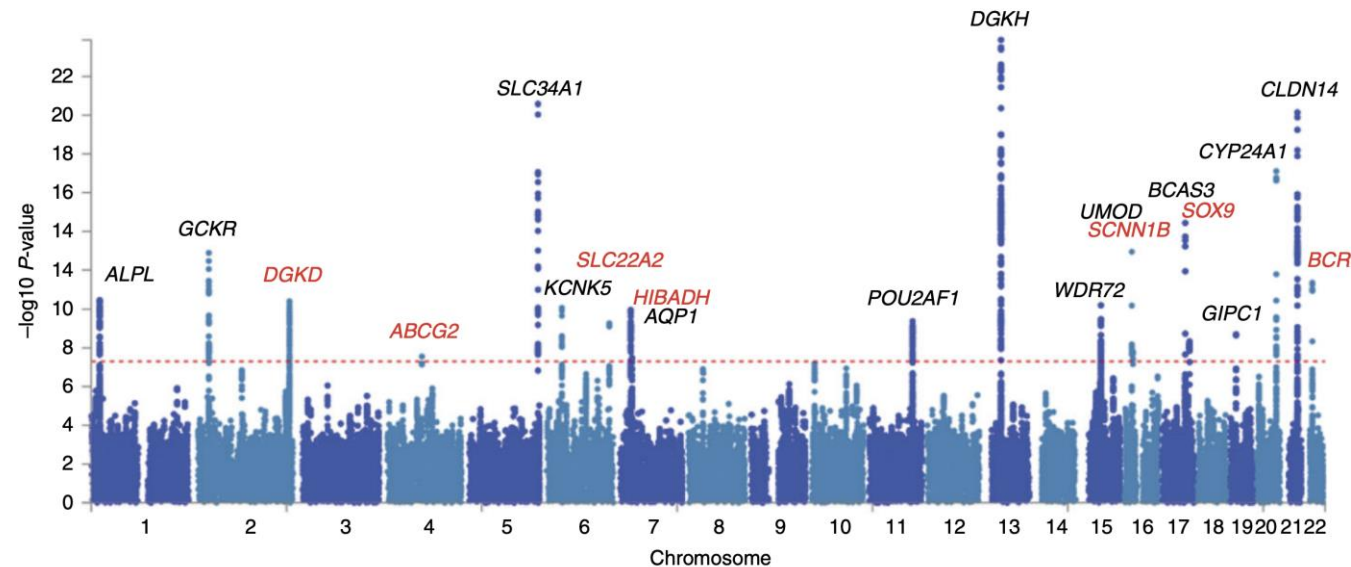
Gap

Existing approaches provide interpretable features, but rarely test every token in a unified inferential framework with effect size, direction and p-value

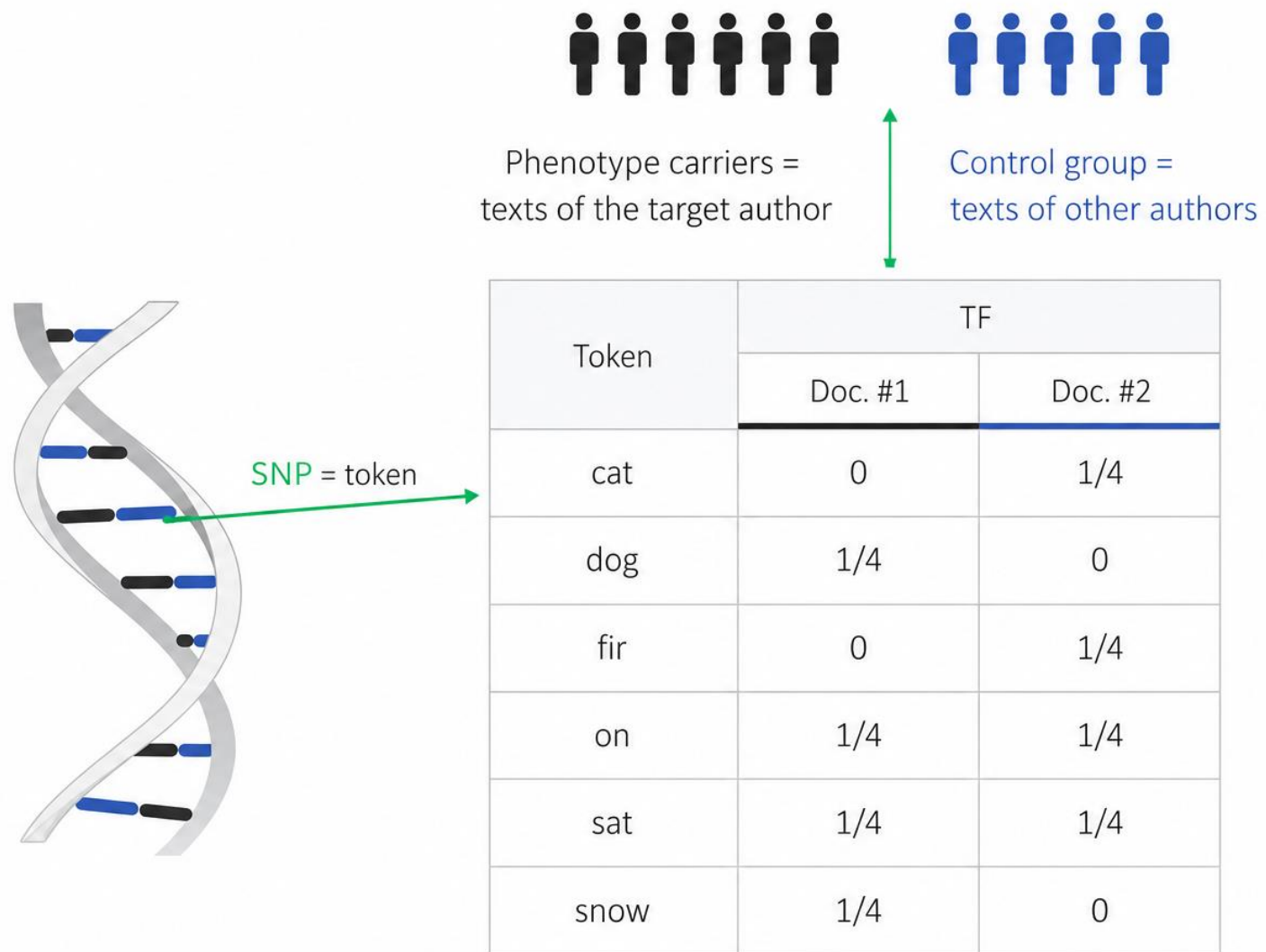
what is GWAS?

A genome-wide association study (abbreviated GWAS) is a research approach used to identify genomic variants that are statistically associated with a particular trait.

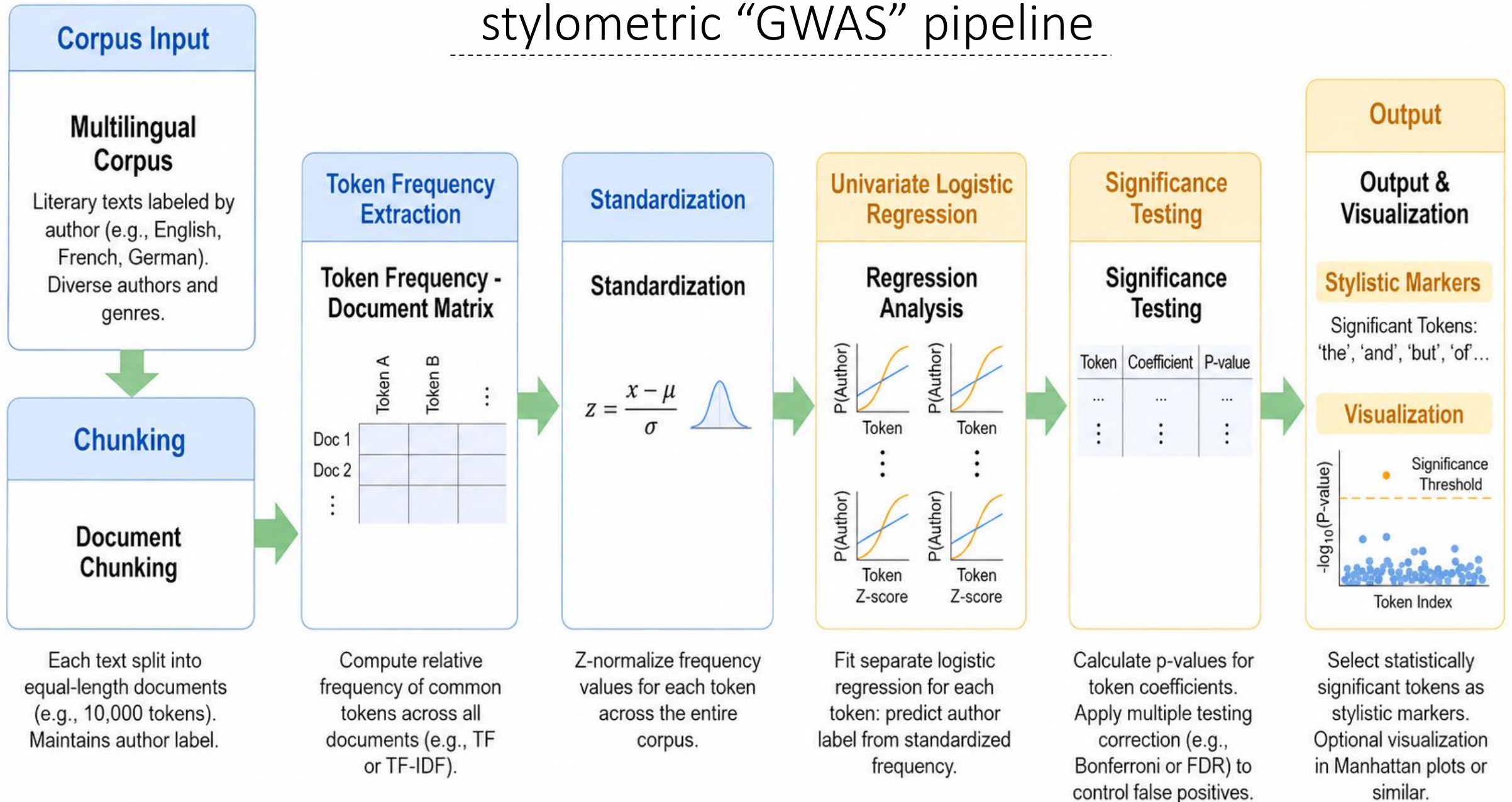
GWAS involves surveying the genomes of many people, looking for genomic variants that occur more frequently in those with a specific trait compared to those without the trait.



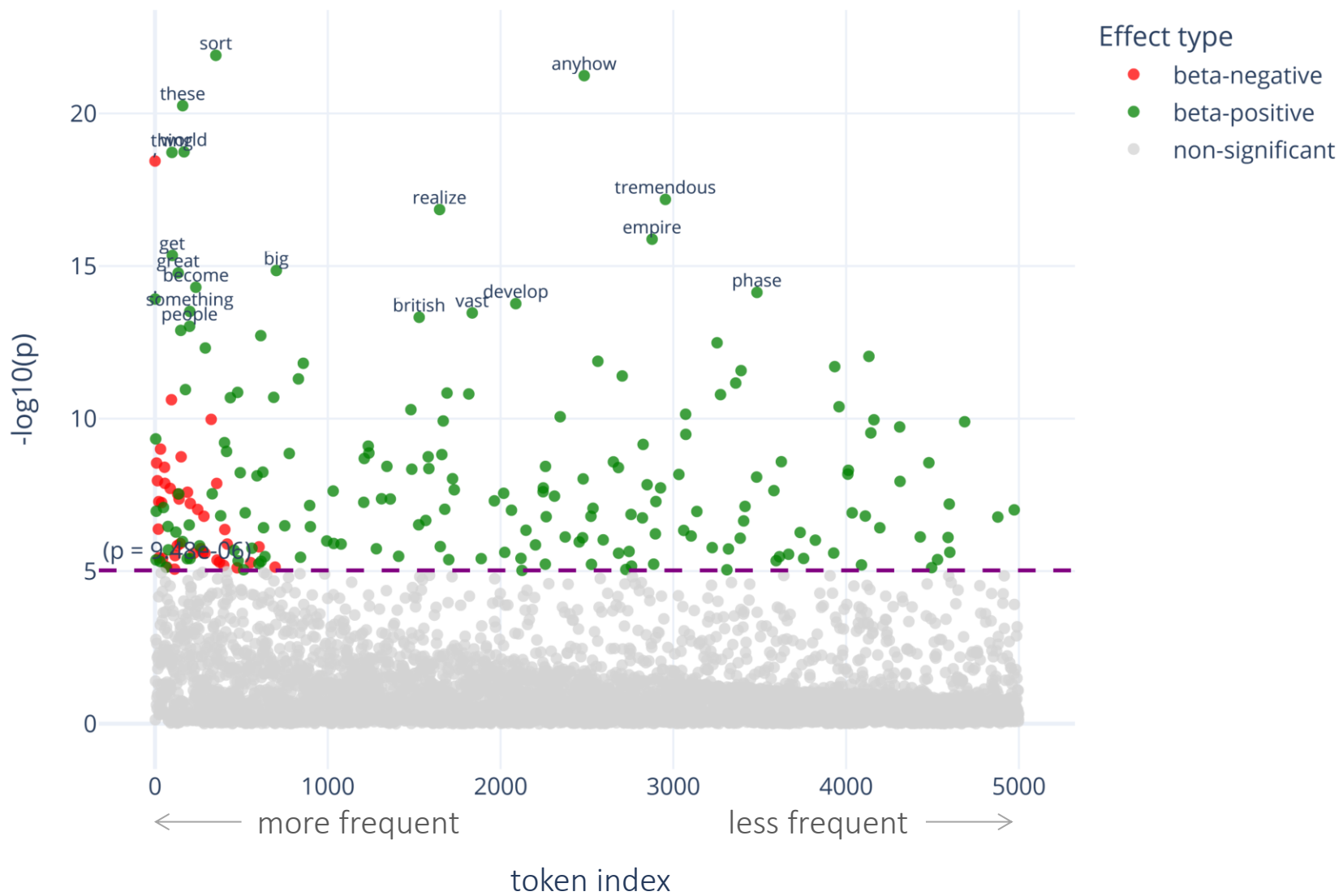
stylometric “GWAS”



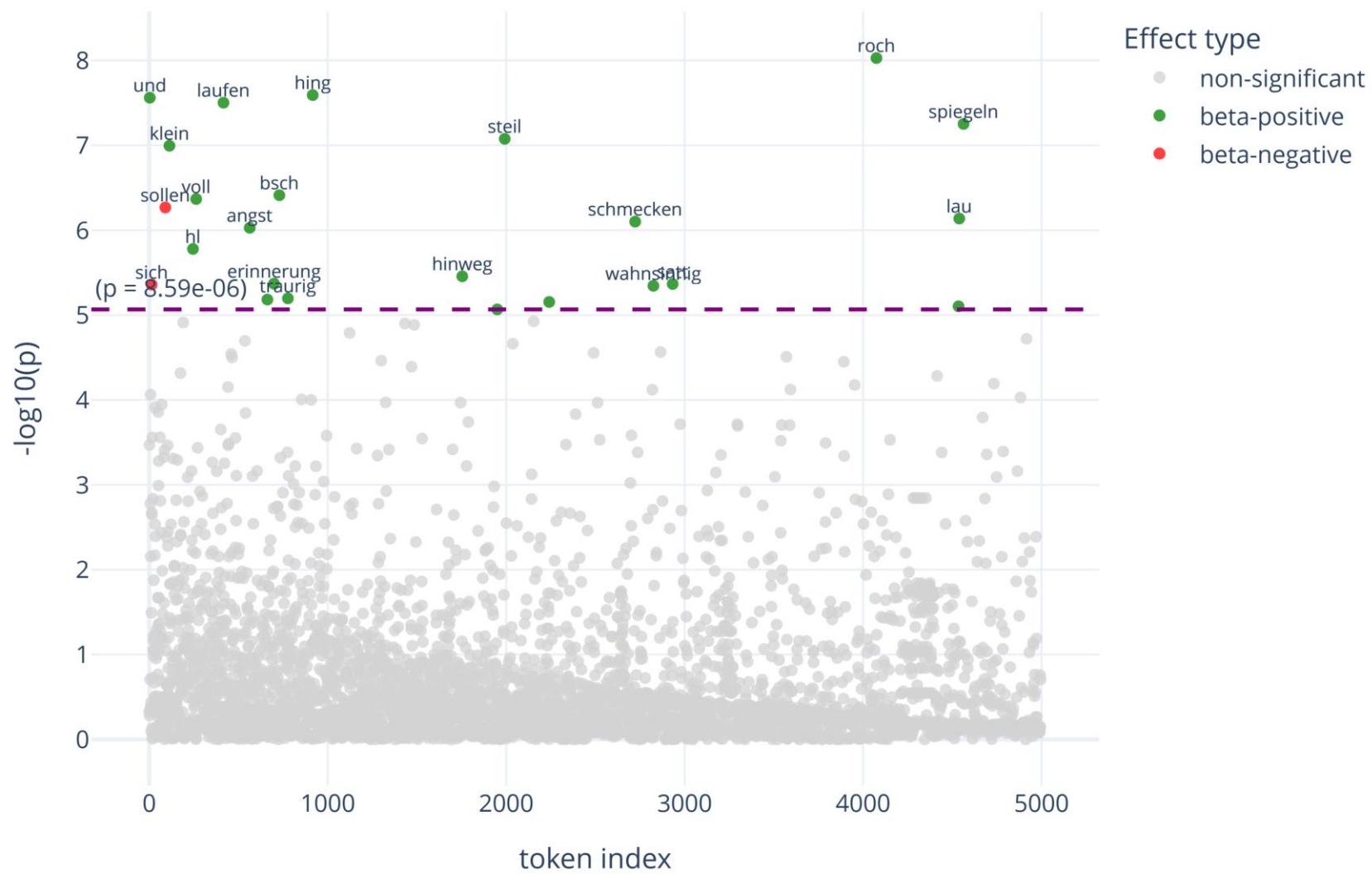
stylometric “GWAS” pipeline



Manhattan plot

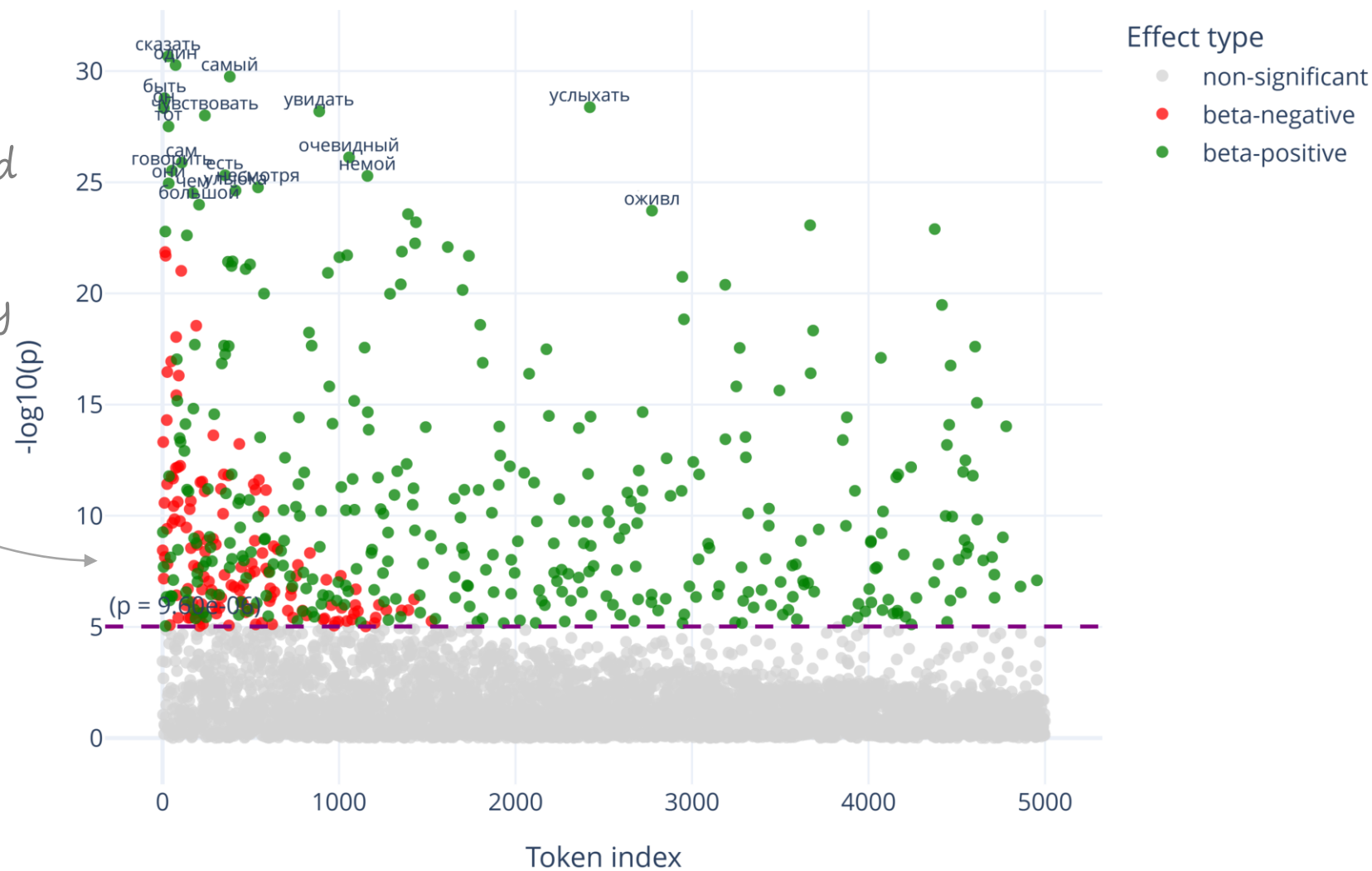


Manhattan plot



L. Tolstoy

Manhattan plot



Further Development

Linguistically informed visualization

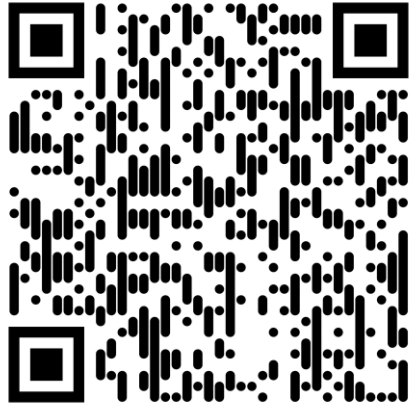
Significant tokens can be grouped on the plot by linguistic categories (like chromosomes in GWAS), such as parts of speech, semantic fields, or functional classes.

Covariate-aware models

Future versions may include genre, period, corpus source, and text length as covariates to separate authorial markers from contextual effects.

Beyond binary comparison

The framework can be extended from one-vs-rest testing to multi-author and mixed-effects models.



Thank you for attention!

Links and contacts

Corresponded author's email: i@ddpronin.ru

Short paper preprint: <https://arxiv.org/abs/2606.09543>

Code: github.com/DDPronin/GWAS-stylometry

